Lot Title : Supply, Installation, Testing, Commissioning and Operation & Maintenance of 400 kw On grid Solar PV System

Bid Security : 1200000 PKR

Job	Delivery Schedule	Quantity
Civil Works	**Address:** Comsats University Islamabad, University Road, near Officer Colony, Attock, Attock, Attock (District), Rawalpindi Division (Division), Punjab (Province). **Schedule:** 180 Days **Quantity:** 1/Qty	1/Qty
Electrical Works	**Address:** Comsats University Islamabad, University Road, near Officer Colony, Attock, Attock, Attock (District), Rawalpindi Division (Division), Punjab (Province). **Schedule:** 180 Days **Quantity:** 1/Qty	1/Qty
Installation, Testing Commissioning, O&M Services and Training	**Address:** Comsats University Islamabad, University Road, near Officer Colony, Attock, Attock, Attock (District), Rawalpindi Division (Division), Punjab (Province). **Schedule:** 180 Days **Quantity:** 1/Qty	1/Qty

Related Services :

Yes

Supply, Installation, Testing, Commissioning and Operation & Maintenance of 400 kw On grid Solar PV System

Job	Related Services
Civil Works	Site clearance and removal of debris
Electrical Works	Site clearance and removal of debris
Installation, Testing Commissioning, O&M Services and Training	Site clearance and removal of debris

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Work Specifications and Market Rates

Lot Title: Supply, Installation, Testing, Commissioning and Operation & Maintenance of 400 kw On grid Solar PV System

Specifications / Requirements:

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
1	Civil Works						

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
2	PV Mount Structure for a) Computer Science Block(150kw) b) Math’s/Management Science Block(150kw) c) Electrical & Computer Engineer Block(100kw	Supply and fixing PV Module Site preparation works including clearance and dismantling of excavation/filling/compaction. Wind speed = 160km/h, azimuth as per site. (UV resistant,120 micron of zinc coating), 12 SWG i.c 2 coat of enamel paint, and one coat of red oxide of approved color or as per drawing given in Employer’s Design. complete in all respect. Or as directed by engineer Incharge.	Sq.M	2250	6000	Six Thousand Only	13500000

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
3	Cable Tray	Supply and fixing of 16 SWG Galvanized iron with flat cover, perforated downside, complete fitting accessories i.e nut bolts, supporting angle iron etc, size (W*D) inch 8*4 for laying string wires or as per International Standard or complete in all respect.	M	200	2250	Two Thousand Two Hundred Fifty only	450000
4	UPVC pipe	Supply and fixing of 75mm dia pipe B class for laying string wires PV area to inverters including sockets, i/c excavation, backfilling complete in all respect. Or as directed by engineer incharge.	M	950	600	Six Hundred Only	570000

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
5	UPVC pipe	50mm dia pipe B class for laying string wires PV area to inverters including socketsi/c excavation, backfilling complete in all 5respect. Or as directed by engineer incharge.	M	700	400	Four Hundred Only	280000
6	UPVC pipe	25mm dia pipe B class for laying string wires PV area to inverters including socketsi/c excavation, backfilling complete in all respect. Or as directed by engineer Incharge.	M	500	200	Two Hundred Only	100000

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
7	PCC(1:3:6)	Clearing, grabbing, Dressing, leveling, compacting and Providing and laying 1:3:6 (1 cement 3 Lawrencepur sand and 6 coarse aggregate crushed graded boulders (machine broken) are used) cement concrete using graded gravel (bajri) 1inch (75 mm) and down gauge in foundation including levelling, compacting and curing etc. complete in all respect. Or as directed by engineer Incharge.	Cubic M	210	1772	One Thousand Seven Hundred Seventy Two Only	372000

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
8	Tuff Paver	Providing and laying in floor C.C. 1:3:6: tuff paver 8"x4"x2-1/2" thick of approved design and colour pattren (average strength 7000 psi) laid on sand cushion filling of joint with sand and warring etc. complete as per direction of Engineer Incharge. (the cost of sand cushion is included)	Sq.M	2250	124.45	One Hundred Twenty Four Only	280000
9	Electrical Works						
10	PV module	Tier 1 Manufacturer according to Bloomberg latest issued rankings with IEC 61215 Compliant, not less than 650 W N- Type and Half Cut PV modules or better	KW	400	39000	Thirty Nine Thousand Only	15600000
11	Hybrid Inverter 150kw 150kw 100kw	3 phase On grid Inverter with Remote Monitoring Devices	KW	400	13000	Thirteen Thousand Only	5200000

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
12	Communication Box	Data Logger complete in all respect as per international standard or NPERA/IESCO.	No.	As per inverter Configuration			400000
13	Genset Synchronization Module	Genset Auto Synchronization Hybrid Genet Controller/Solar-Diesel Integration Controller compatible with Inverter and Generator, factory fabricated as approved by the Engineer.	Nos.	3	250000	Two Hundred Fifty Thousand Only	750000
14	Lightning Arrester	Copper rod height 2 m to be fixed on these five Site on required height with poles complete in all respect as per drawing etc as per international standard safety or NPERA/IESCO.	Nos.	5	100000	One Hundred Thousand Only	500000
15	0.4kV LV Panel (1x1500 Kw,1x150kW & 1x100kW)	Details of breakers and SPD to be in housed in cabinet: for AC and DC System As given in AC & DC BoQ and drawings attached	Nos.	3	210000	Two Hundred Ten Thousand Only	630000

PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
16	DC Distribution Box	IP65 GI DB for each inverter for placing DC strings Expected size but not limited to W*H*D = 3*1.5*0.75(16SWG) wall mount, single door with lock and with complete fitting scheme/ railing for DC Breakers	Nos.	3	50000	Fifty Thousand Only	150000
17	DC Breakers	As per Hybrid configuration and PV Module fuse ratings, as per NEPRA and International Standard or complete in all respect.	Nos.	As per inverter Hybrid Configuration			210000
18	DC wire for PV	6 mm Sq. DC flexible wire 1500V DC Copper conductor, Cu XLPO for array to DB(Double Insulated)	M	7000	380	Three Hundred Eighty Only	2660000

PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
19	DC wire for PV	25 mm Sq. DC flexible wire 1500V DC Copper conductor, Cu XLPO from array DB to Main DB(Double Insulated)	M	250	2200	Two Thousand Two Hundred Only	550000
20	DC wire for PV	16 mm Sq. DC flexible wire 1500V DC Copper conductor, Cu XLPO from array DB to Main DB(Double Insulated)	M	250	1700	One Thousand Seven Hundred Only	425000
21	AC wire from LV panel to Inverter	Cu, XLPE/PVC 0.6/1 kV, 4*120 mm Sq. AC stranded(Double Insulated)	M	120	15000	Fifteen Thousand Only	1800000
22	AC wire for inverter grounding	Cu, PVC 0.6/1 kV, 1*70 mm Sq. AC flexible wire stranded	M	150	1500	Fifteen Hundred Only	225000
23	DC wire for PV panel and PV frames to grounding Point	PVC 1*16 mm Sq. AC standard Wire	M	600	650	Six Fifty Only	390000

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
24	Copper Bare Conductor	25 mm Sq	M	100	980	Nine Hundred Eighty Only	98000
25	MC4 connectors	IP67/IP68 for PV Hybriding as per International Standard complete in all respect.	Nos.	200	260	Two Hundred Sixty Only	52000
26	Bi-directional Meter for Net Metering	Delivery of Bi-directional Meter in line with NEPRA / IESCO requirements	Nos.	3	140000	One hundred Forty Thousand Ony	420000
27	DC Earthing	Bore Type Earthing for DC Components as Per NEPRA or international Standard of safety	Nos.	3	64900	Sixty Four Thousand Nine Hundred Only	194700

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
28	AC Earthing	Bore Type Earthing for AC Components as Per NEPRA Standard or international Standard of safety	Nos.	6	156000	One Hundred Fifty Six Thousand Only	468000
29	Motorized Panel	• 6 x Motorized Panel Cleaners (battery operated) with minimum7-meter pole length complete in all respect.	1	Job	194700	One Hundred Ninty Four Thousand Seven Hundred Only	194700
30	Installation, Testing and Commissioning O&M Services and Training						
31	Installation, Testing and Commissioning	Complete project installation and commissioning till handover of the Project to Employer	Job/Nos	3	400000	Four Hundred Thousand Only	1200000

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
32	Net billing services	Net Billing services complete in all aspects including NEPRA license and load flow studies required by the distribution company.	Job/No’s	3	500000	Five Hundred Thousand Only	1500000
33	Operation and Maintenance Services	Cleaning of all PV Modules on each Saturday / Sunday for 06 months. Monitoring of System Performance and submission of monthly electricity generation reports	Job/ No’s	2	135000	One Hundred Thirty Five Thousand Only	270000
34	O&M Training	1 week O & M training of 3 personnel of the Employer	Job.	1	50000	Fifty Thousand Only	50000

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Sr. No	Ref No.	Description	Unit	Qty	Rate		Amount
35	MISC ITEMS	• Cable ties, rawal bolts, pvc tapes, steel and PVC clamps, pvc pipes, glands different sizes, thimbles, connectors, warning labels etc. • 2 x insulated glove pairs, shoe pairs, hats for site inspection and maintenance activities • PV module O&M Toolkit 1 x 55 inch 4K TV Screen for monitoring of generation data from all inverters complete in all respect.	Job	1	510600	Three Hundred Twenty Thousand Only	510600
						GRAND TOTAL:	Rs50,000,000.00

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Scope of Work

Solar PV Modules

IEC 61215 Certified
Application/ Safety Class II as per IEC 61730
N-Type Half Cut Cell
Tier 1 Manufacturer as per the most recent Bloomberg list, AAA or AA rated as per PV Module-Tech Ratings
650 W or higher with positive tolerance only
23% or higher module efficiency
25 Year Linear Performance Warranty
Anti-Reflective Coating

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

12 minimum Year Material and replacement warranty of panel
Power Output after 25 Years ≥ 85%
Potential Induced Discharge(PID) Resistant
Light Induced Degradation (LID) equal to 0.7%
Module Temperature Coefficient at Pmax of -0.30%/oC or better
Operating Temperature Range -40 to 85 oC
Barcode Identification
Efficiency at 200 W/m2≥ 96% of STC
Report of Real Power at STC measured with AAA Class Sun Simulator at the factory to be provided
Compliance with S.R.O 604 defined Pre-shipment inspection requirements as per Govt of Pakistan

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Solar PV Mounting Structure

UV resistant, Hot Dip Galvanized Steel frame of Gauge 12 with 120 micron or thicker zinc coating or better (More details under the drawing attached)
Tilt 25o to 33o Degrees, Azimuth 290 w.r.t North to be determined from shadow analysis
Shall withstand wind speeds of 160 km/h
Galvanic Isolation
PV module fastening: As per module manufacturer guidelines
Rawal bolts: Stainless Steel (Grade-304) accompanied with epoxy resin backfilling for water proofing
Inverters& Hybrid Formation

Three Phase Grid Tied
Hybrid Inverter

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Hybrids formation priority: 1.Same Row,2.Adjacent Row
Hybrid Voc: Must be within MPPT range at -10 to + 80 Degree Celsius cell temperature
Anti-Potential Induced Degradation Module (PID)
Sheds and enclosures for outdoor installation
50 to 60 Hz Frequency Output
CEC Efficiency of 95.5% or above
IP66 rated
12 Year Plus Warranty
THD 3%
IEC 62109/UL1741 Compliant Safety Design

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Transformer-less modular Design
Ambient Operating Temperature Range: -25 oC to 60 oC
Multiple MPPT Trackers
DC/AC ratio not more than 1.1
Residual Current Monitoring
DC Insulation Resistance Detection
Generator Synchronization module
AC and DC Surge Arrestors: Type II or better or certified per IEC 61643-11 or 20 KA for AC and DC or equivalents
AC Overcurrent Protection
DC reverse polarity protection

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

PV Array Hybrid Fault Monitoring
Anti-Islanding Protection
WiFi and Cloud Communication
USB, Modbus, RS485
S.R.O 604 Compliant

AC,DC Cables, Earthing wires &Cable Trays

Copper Cables with purity of 99.99%
UL 4703 compliant Low Smoke Halogen Free
DC cable size : 06 mm2 from array of panel to DB DC cable size : 16 mm2 from array DB to Main DB AC cable size : As per attached Drawing

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

XLPO Cables for operation under elevated temperatures rated for 1500 V
Conduits shall have 30% or higher spare capacity
Preparation of Site and all Civil Works
Seamless Terminations
Perforated Cable Trays of 16 Gauge or thicker
Avoid Direct Sunlight and Direct Clamping of Cables

Cables should not be physically accessible for routine works
Labeling of all Hybidsequipment shall be mandatory
06 mm2 earth wire for equipotential bonding

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Cables to be laid in UPVC/corrugated PVC pipes and cable trays

System Design and Installation/As Built Drawings

Energy Yield Assessment
Module Layout Drawings as per the drawings attached and Referred
Cable Layout Plans
As built Drawings

Breakers, Surge Arrester & LV Panel

Circuit Breakers
Changeover Switches

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Class-II Surge protection separate for AC & DC sides
Separate DC and AC Earthing each with 5 Ohms or lesser resistance
Non-Isolated Lightning Arrestors
Protection against lightning currents of 31 kA or higher
In case of LV panel installation, the panel shall be IP65 rated with galvanized material of at least 16 gauge with protection Scheme as per IEC Standard 62548, having Input and Output Circuit Breakers as per IEC Standards, Bus Bar Current Bearing Capacity as per requirement, Isolators for Bus Bars and AC Grounding Bus Bar.
The panel shall have Digital Ammeter and Voltmeter, RYB indicator with Breather Vents on both Sides of cabinets with RAL Color: 6021

Monitoring and Control System

Centralized Monitoring and Logging of array level input, System Output

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Communication through the following:

1. RS485/RS232/TTL/USB for onsite data collection/monitoring
2. GPRS/4G for real-time data transfer to Client's defined server (subscription of

GPRS/4G to be paid by the Contractor)

1. Wifi/Bluetooth Module for localized wireless communication
2. Alarm Messages (SMS & Email) for Fault/ Grid Outage / Generator Outage / Inverter Outage etc.
3. Daily Generation data
4. Each Inverter data
5. Monthly production data
6. Yearly production data
7. Total production data

Net Metering Application and Installation of Bidirectional Meter

Compliance with all prevailing regulations set forth by NEPRA/IESCO/Punjab Energy Department for net metering and safety purposes

Load flow studies
Preparation and submission of application for net metering

Specific Plant Data For 03xAcademic Block At CUI Attock Campus (400 kW)

Operating status
Vdc
Vac
Pdc
Pac
Grid Status
Generator Status (from Genset Sync. Module)

PUBLIC PROCUREMENT REGULATORY AUTHORITY (PPRA) EPADS

Fault Detection
Daily Energy Supplied by PV System
Total Energy Supplied by PV System
Plant Performance Data Check: Every 15 Seconds or lesser
Total Energy Supplied by Genset (from Genset Sync. Module)

Synchronization Modules for GENSETS AND

INVERTERS

GENSETS syncing modules Compatible with Diesel generators and Proposed Inverters

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Price Schedule

For Individual Jobs

#	Job Title	Quantity	Unit Price (PKR)	Total Price (PKR)	Delivery Location	Delivery Period / Year	Country of Origin
1							
2							

For Lots

#	Lot Title	Total Lot Price (PKR)	Country of Origin
1	[Lot 1 Title]		

PUBLIC PROCUREMENT REGULATORY AUTHORITY
(PPRA)
EPADS

General Conditions of Contract

A. General

1. **Definitions**

1.1. Unless the context otherwise requires, the following terms whenever used in this Contract shall have the same meaning and shall be interpreted as indicated

1.1.1. “Applicable Law” means the laws and any other instruments having the force of law in the Government’s Country, or in such other country as may be specified in the Special Conditions of the Contract (SC), as they may be issued and in force from time to time;

1.1.2. “The Contract” means an agreement enforceable by law;

1.1.3. “The Contract Price” means the price payable to the Contractor under the Contract for the full and proper performance of its contractual obligations;

1.1.4. “The Services” means the work to be performed by the Contractor pursuant to this Contract and as prescribed in the Specifications and Schedule of Activities included in the Contractor’s Bid;

1.1.5. “Ancillary Services” means those services ancillary to the provision of Services, such as transportation and insurance, and any other incidental services, such as installation, commissioning, provision of technical assistance, training, and other such obligations of the Contractor covered under the Contract;

1.1.6. “GCC” means the General Conditions of Contract contained in this section;

1.1.7. “SCC” means the Special Conditions of Contract by which the GCC may be amended or supplemented;

1.1.8. **“**Day” means calendar day unless indicated otherwise;

1.1.9. “Effective Date” means the date on which this Contract comes into force and effect;

1.1.10. “The Contractor” means the individual or corporate body whose Bids to provide the Services has been accepted by the Procuring Agency;

1.1.11. “The Project Site,” where applicable, means the place or places named in Bid Data Sheet and technical Specifications;

1.1.12. “Government” means the Government of Pakistan;

1.1.13. “Local Currency” means the currency of Pakistan;

1.1.14. “In Writing” means communicated in written form with proof of receipt;

1.1.15. “Completion Date” means the date of completion of the Services by the Contractor as certified by the Procuring Agency;

1.1.16. “Foreign Currency” means any currency other than the currency of the country of the Procuring Agency;

1.1.17. “Party” means the Procuring Agency or the Contractor, as the case may be, and “Parties” means both of them;

1.1.18. "Service" means any object of procurement other than goods or works;

1.1.19. “Subcontractor” means any entity to which the Bidder subcontracts any part of the Services.

2. **Applicable Law**

2.1. The contract shall be governed and interpreted in accordance with the laws of Pakistan, unless otherwise specified in SCC.

3. **Language**

3.1. The Contract as well as all correspondence and documents relating to the Contract exchanged between the Contractorand the Procuring Agency, shall be written in the **English language** unless otherwise stated in the **SCC** . Supporting documents and printed literature that are part of the Contract may be in another language provided these are accompanied by an accurate translation of the relevant passages in English, in which case, for purposes of interpretation of the Contract, this translation shall govern.

4. **Notices**

4.1. Any notice, request, or consent made pursuant to this Contract shall be in writing and shall be deemed to have been made when delivered in person to an authorized representative of the Party to whom the communication is addressed, or when sent by registered mail, telex, telegram, or facsimile to such Party at the address specified in the **SCC**.

5. **Location**

5.1. The Services shall be performed at such locations as the Procuring Agency may approve and as specified in **SCC.**

5.2. A {DOCUMENTS}

6. **Authorized Representatives / Authority of Member in charge**

6.1. Any action required or permitted to be taken, and any document required or permitted to be executed, under this Contract by the Procuring Agency or the Contractor may be taken or executed by the officials specified in the **SCC**.

B. Commencement, Completion, Modification, and Termination of Contract

7. **Effectiveness of Contract**

7.1. This Contract shall come into effect on the date the Contract is signed by both parties and such other later date as may be stated in the **SCC.**

8. **Commencement of Services**

8.1. The Contractor shall confirm availability of Key Experts and begin carrying out the Services not later than the number of days after the Effective Date specified in the **SCC.**

9. **Program schedule**

9.1. Before commencement of the Services, the Contractor shall submit to the Procuring Agency for approval a Program showing the general methods, arrangements, order and timing for all activities. The Services shall be carried out in accordance with the approved Program as updated.

10. **Starting Date/Expiration Date**

10.1. The Contractor shall start carrying out the Services Five (05) days after the date the Contract becomes effective, or at such other date as may be specified in the **SCC.**

10.2. Unless terminated earlier pursuant to Clause **GCC 14** hereof, this Contract shall expire at the end of such time period after the Effective Date as specified in the **SCC.**

11. **Entire Agreement**

11.1. This Contract contains all covenants, stipulations and provisions agreed by the Parties. No agent or representative of either Party has authority to make, and the Parties shall not be bound by or be liable for, any statement, representation, promise or agreement not set forth herein.

12. **Modification**

12.1. Any modification or variation of the terms and conditions of this Contract, including any modification or variation of the scope of the Services, may only be made by written agreement between the Parties. However, each Party shall give due consideration to any modification(s) or variation(s) made by the other Party.

12.2. In cases of any modification(s) or variation(s), the prior written consent of the Procuring Agency is required.

13. **Force Majeure**

13.1. **Definition**
For the purposes of this Contract, “Force Majeure” means an event which is beyond the reasonable control of a Contractor and which makes a Contractor’s performance of its obligations under the Contract impossible or so impractical as to be considered impossible under the circumstances.

13.2. **No Breach of Contract**
The failure of a Party to fulfill any of its obligations under the contract shall not be considered to be a breach of, or default under, this Contract in so far as such inability arises from an event of Force Majeure, provided that the Party affected by such an event (a) has taken all reasonable precautions, due care and reasonable alternative

measures in order to carry out the terms and conditions of this Contract, and (b) has informed the other Party as soon as possible about the occurrence of such an event.

13.3. **Extension of Time**
Any period within which a Contractor shall, pursuant to this Contract, complete any action or task, shall be extended for a period equal to the time during which such Party was unable to perform such action as a result of Force Majeure.

13.4. **Payments**
During the period of their inability to perform the Services as a result of an event of Force Majeure, the Contractor shall be entitled to continue to be paid under the terms of this Contract, as well as to be reimbursed for additional costs reasonably and necessarily incurred by them during such period for the purposes of the Services and in reactivating the Service after the end of such period.

14. **Termination**

14.1. **By the Procuring Agency**
The Procuring Agency may terminate this Contract in case of the occurrence of any of the events specified in paragraphs (a) through (e) of this Clause. In such an occurrence the Procuring Agency shall give at least thirty (30) calendar days' written notice of termination to the Contractor in case of the events referred to in (a) through (d); at least sixty (60) calendar days' written notice in case of the event referred to in (e);

14.1.1. If the Contractor fails to remedy a failure in the performance of its obligations hereunder, as specified in a notice of suspension;

14.1.2. If the Contractor becomes (or, if the Contractor consists of more than one entity, if any of its members becomes) insolvent or bankrupt or enter into any agreements with their creditors for relief of debt or take advantage of any law for the benefit of debtors or go into liquidation or receivership whether compulsory or voluntary;

14.1.3. If the Contractor fails to comply with any final decision reached as a result of arbitration proceedings;

14.1.4. If, as the result of Force Majeure, the Contractor is unable to perform a material portion of the Services for a period of not less than sixty (60) calendar days;

14.1.5. If the Procuring Agency, in its sole discretion and for any reason whatsoever, decides to terminate this Contract;

14.2. **By the Contractor**
The Contractor may terminate this Contract, by not less than thirty (30) calendar days' written notice to the Procuring Agency, in case of the occurrence of any of the events specified in paragraphs (a) through (d) of this Clause.

14.2.1. If the Procuring Agency fails to pay any money due to the Contractor pursuant to this Contract and not subject to dispute within forty-five (45) calendar days after receiving written notice from the Contractor that such payment is overdue;

14.2.2. If, as the result of Force Majeure, the Contractor is unable to perform a material portion of the Services for a period of not less than sixty (60) calendar days;

14.2.3. If the Procuring Agency fails to comply with any final decision reached as a result of arbitration;

14.2.4. If the Procuring Agency is in material breach of its obligations pursuant to this Contract and has not remedied the same within forty-five (45) days (or such longer period as the Bidder may have subsequently approved in writing) following the receipt by the Procuring Agency of the Contractor's notice specifying such breach.

C. Obligations of the Contractor

15. **General**

15.1. **Standard of Performance**

15.1.1. The Contractor shall perform the Services and carry out the Services with all due diligence, efficiency and economy, in accordance with generally accepted professional standards and practices, and shall observe sound management practices, and employ appropriate technology and safe and effective equipment, machinery, materials and methods. The Contractor shall always act, in respect of any matter relating to this Contract or to the Services, as a faithful adviser to the Procuring Agency, and shall at all times support and safeguard the Procuring Agency's legitimate interests in any dealings with the third parties;

15.1.2. The Contractor shall employ and provide such qualified and experienced Experts and Sub-Contractors as are required to carry out the Services.

15.2. **Law Applicable to Services**

The Contractor shall perform the Services in accordance with the Contract and in accordance with the Law of Pakistan and shall take all practicable steps to ensure that any of its Experts and Sub-Bidders, comply with the Applicable Law.

16. **Conflict of Interests**

16.1. **Contractor Not to Benefit from Commissions and Discounts**

The remuneration of the Contractor shall constitute the Contractor's sole remuneration in connection with this Contract or the Services, and the Contractor shall not accept for their own benefit any trade commission, discount, or similar payment in connection with activities pursuant to this Contract or to the Services or in the discharge of their obligations under the Contract, and the Contractor shall use their best efforts to ensure that the Personnel, any Subcontractors, and agents of either of them similarly shall not receive any such additional remuneration.

16.2. **Contractor and Affiliates Not to be Otherwise Interested in Project**

The Contractor agree that, during the term of this Contract and after its termination, the Contractor and its affiliates, as well as any Subcontractor and any of its affiliates, shall be disqualified from providing Services (other than the Services and any continuation thereof) for any project resulting from or closely related to the Services.

16.3. **Prohibition of Conflicting Activities**

Neither the Bidder nor its Subcontractors nor the Personnel shall engage, either directly or indirectly, in any of the following activities:

16.3.1. during the term of this Contract, any business or professional activities in the Government's country which would conflict with the activities assigned to them under this Contract;

16.3.2. during the term of this Contract, neither the Contractor nor their Subcontractors shall hire public employees in active duty or on any type of leave, to perform any activity under this Contract;

16.3.3. after the termination of this Contract, such other activities as may be specified in the **SCC.**

17. **Insurance to be Taken Out by the Contractor**

17.1. The Contractor(a) shall take out and maintain, and shall cause any Subcontractors to take out and maintain, at its (or the Sub-contractors', as the case may be) own cost but on terms and conditions approved by the Procuring Agency, insurance against the risks, and for the coverage, as shall be specified in the **SCC**; and (b) at the Procuring Agency's request, shall provide evidence to the Procuring Agency showing that such insurance has been taken out and maintained and that the current premiums have been paid.

18. **Contractor's Actions Requiring Procuring Agency's Prior Approval**

18.1. The Contractor shall obtain the Procuring Agency's prior approval in writing before taking any of the following actions:

18.1.1. appointing such members of the Personnel not provided by the Contractor;

18.1.2. changing the Program of activities; and

18.1.3. any other action that may be specified in the **SCC.**

19. **Reporting Obligations**

19.1. The Contractor shall submit to the Procuring Agency the reports and documents in the numbers, and within the periods as prescribed by the Procuring Agency.

20. **Liquidated Damages**

20.1. **Payments of Liquidated Damages**

The Contractor shall pay liquidated damages to the Procuring Agency at the rate per day stated in the **SCC** for each day that the Completion Date is later than the Intended Completion Date. The total amount of liquidated damages shall not exceed the amount defined in the **SCC**. The Procuring Agency may deduct liquidated damages from payments due to the Contractor. Payment of liquidated damages shall not affect the Contractor's liabilities.

20.2. **Correction for Over-payment**

If the Intended Completion Date is extended after liquidated damages have been paid, the Procuring Agency shall correct any overpayment of liquidated damages by the Contractor by adjusting the next payment certificate. The

Contractor shall be paid interest on the overpayment, calculated from the date of payment to the date of repayment, at the rates specified in **SCC.**

20.3. **Lack of performance penalty**
If the Contractor has not corrected a Defect within the time specified in the Procuring Agency's notice, a penalty for Lack of performance will be paid by the Contractor. The amount to be paid will be calculated as a percentage of the cost of having the Defect corrected, assessed as specified in the Contractor

21. **Performance Guarantee**

21.1. Within the time stipulated in the acceptance letter from the Procuring Agency, the successful Bidder shall furnish the Performance Guarantee in shape and amount **specified in SCC.**

21.2. The proceeds of the Performance Guarantee shall be payable to the Procuring agency as compensation for any loss resulting from the Supplier's failure to complete its obligations under the Contract.

21.3. The Performance Guarantee shall be denominated in the currency of the Contract, or in a freely convertible currency acceptable to the Procuring agency and shall be in the acceptable form as specified in **SCC**.

21.4. The Performance Guarantee will be discharged by the Procuring agency and returned to the Supplier not later than thirty (30) days following the date of completion of the Supplier's performance obligations under the Contract, including any warranty obligations, unless otherwise **specified in SCC.**

22. **Sustainable Procurement**

22.1. The Contractor shall conform to the sustainable procurement contractual provisions, if and as specified in the **SCC.**

D. Contractor's Personnel

23. **Description of Personnel**

23.1. The titles, agreed job descriptions, minimum qualifications, and estimated periods of engagement in the carrying out of the Services of the Contractor's Key Personnel. The Key Personnel listed by title as well as by name are hereby approved by the Procuring Agency.

24. **Removal and / or Replacement of Personnel**

24.1. Except as the Procuring Agency may otherwise agree, no changes shall be made in the Key Personnel. If, for any reason beyond the reasonable control of the Contractor, it becomes necessary to replace any of the Key Personnel, the Contractor shall provide as a replacement a person of equivalent or better qualifications.

24.2. If the Procuring Agency finds that any of the Personnel have (i) committed serious misconduct or have been charged with having committed a criminal action, or (ii) have reasonable cause to be dissatisfied with the performance of any of the Personnel, then the Contractor shall, at the Procuring Agency's written request specifying the grounds thereof, provide as a replacement a person with qualifications and experience acceptable to the Procuring Agency.

24.3. The Contractor shall have no claim for additional costs arising out of or incidental to any removal and/or replacement of Personnel.

E. Obligations of the Procuring Agency

25. **Change in the Applicable Law**

25.1. If, after the date of this Contract, there is any change in the Applicable Law with respect to taxes and duties which increases or decreases the cost of the Services rendered by the Contractor, then the remuneration and reimbursable expenses otherwise payable to the Contractor under this Contract shall be increased or decreased accordingly by agreement between the Parties, and corresponding adjustments shall be made to the amounts referred in the **SCC.**

26. **Services and Facilities**

26.1. The Procuring Agency shall make available to the Contractor and the Experts, for the purposes of the Services and free of any charge, the services, facilities and property described in the Terms of Reference, at the times and in the manner specified in the Terms of Reference.

26.2. In case that such services, facilities and property shall not be made available to the Contractor, the Parties shall agree on (i) any time extension that it may be appropriate to grant to the Contractor for the performance of the Services, (ii) the manner in which the Contractor shall procure any such services, facilities and property from other sources, and (iii) the additional payments, if any, to be made to the Contractor as a result thereof.

F. Payments to the Contractor

27. **Contract Price**

27.1. The price payable shall be in Pakistani Rupees unless otherwise specified in the **SCC**.

28. **Terms and Conditions of Payment**

28.1. Payments will be made to the Contractor according to the payment schedule stated in the **SCC** and as per actual invoice submitted by the Contractor.

28.2. Unless otherwise stated in the **SCC**, the advance payment shall be made against the provision by the Contractor of a bank guarantee for the same amount, and shall be valid for the period stated in the **SCC.** Any other payment shall be made after the conditions listed in the **SCC** for such payment have been met, and the Contractor have submitted an invoice to the Procuring Agency specifying the amount due.

29. **Quality Control Identifying Defects**

29.1. The principle and modalities of Inspection of the Services by the Procuring Agency shall be as indicated in the **SCC.** The Procuring Agency shall check the Contractor’s performance and notify him of any Defects that are

found. Such checking shall not affect the Contractor's responsibilities. The Procuring Agency may instruct the Contractor to search for a Defect and to uncover and test any service that the Procuring Agency considers may have a Defect. Defect Liability Period is as defined in the **SCC**.

29.2. A {INSPECTION}

30. **Correction of Defects, and Lack of Performance Penalty**

30.1. The Procuring Agency shall give notice to the contractor of any Defects before the end of the Contract. The Defects liability period shall be extended for as long as Defects remain to be corrected.

30.2. Every time notice a Defect is given; the contractor shall correct the notified Defect within the length of time specified by the Procuring Agency's notice.

30.3. If the contractor has not corrected a Defect within the time specified in the Procuring Agency's notice, the Procuring Agency will assess the cost of having the Defect corrected, the contractor will pay this amount, and a Penalty for Lack of Performance.

31. **Settlement of Disputes Amicable Settlement**

31.1. The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with this Contract or its interpretation.

32. **Dispute Settlement**

32.1. **Arbitration**

If any dispute of any kind whatsoever shall arise between the procuring agency and the contractor in connection with or arising out of the Contract, including without prejudice to the generality of the foregoing, any question regarding its existence, validity or termination, or the execution of the contract, the parties shall seek to resolve any such dispute or difference by mutual consultation. If the parties fail to resolve such a dispute or difference even after negotiations or mediation, then the dispute shall be referred within fourteen (14) days in writing by either party to the Arbitrator, with a copy to the other party.

Any dispute in respect of which a notice of intention to commence arbitration has been given, in accordance with **GCC sub-clause 32.1**, shall be finally settled by arbitration. Arbitration may be commenced prior to or after completion of the Contract. Arbitration proceedings shall be conducted in accordance with Arbitration Act 1940. Notwithstanding any reference to arbitration herein, the parties shall continue to perform their respective obligations under the Contract unless otherwise agreed. The Procuring Agency shall continue to pay the Contractor any undisputed amounts due under the Contract during the resolution of any dispute.

Special Conditions of Contract

SECTION VIII. SPECIAL CONDITIONS OF CONTRACT

The following Special Conditions of Contract shall supplement the General Conditions of Contract. Whenever there is a conflict, the provisions herein shall prevail over those in the Conditions of Contract. The corresponding clause number of the GCC is indicated in parentheses.

Number of GC Clause

Amendments of, and Supplements to, Clauses in the General Conditions of Contract

Number of GC Clause 1

Definitions

The Procuring Agency is: Comsats University, Islamabad (Attock Campus) (Comsats University, Islamabad), Senior Program Officer Comsats University Islamabad, University Road, near Officer Colony, Attock, Attock, Attock (District), Rawalpindi Division (Division), Punjab (Province).

The Supplier is:

The title of the subject procurement is: Supply, Installation, Testing, Commissioning and Operation & Maintenance of 400 kw On grid Solar PV System

Number of GC Clause 2

Applicable/Governing Law:

The Contract shall be interpreted in accordance with the laws of Islamic Republic of Pakistan

Number of GC Clause 3

Language:

The language of the Contract, all correspondence and communications to be given, and all other documentation to be prepared and supplied under the Contract shall be in **English.**

Number of GC Clause 4

Notices:

The addresses for the notices are:

Procuring Agency:

Comsats University, Islamabad (Attock Campus) (Comsats University, Islamabad), Senior Program Officer Comsats University Islamabad, University Road, near Officer Colony, Attock, Attock, Attock (District), Rawalpindi Division (Division), Punjab (Province).
+92-312-830-9939
ilyas@cuiatk.edu.pk

Contractor/ Bidder:

[Name, address and telephone number]

The Contractor/ Bidder's Representative(s)

[Name, address, telephone number and e-mail address]

Number of GC Clause 6.1

The Authorized Representatives are:

For the Procuring Agency:

Comsats University, Islamabad (Attock Campus) (Comsats University, Islamabad), Senior Program Officer Comsats University Islamabad, University Road, near Officer Colony, Attock, Attock, Attock (District), Rawalpindi Division (Division), Punjab (Province).
+92-312-830-9939
ilyas@cuiatk.edu.pk

For the Bidder:

Name: ……

Designation: ……..

Address: ……..

Number of GC Clause 7

Effectiveness of the contract

The Contractor/Bidder shall be effective within ….. days from the date of signature of the Contract by both parties

Number of GC Clause 8

Commencement of Contract:

The Contractor/Bidder shall provide Non-Consultancy Services from the effective date of contract.

Number of GC Clause 10.2

Expiration of Contract:

The time period shall be ……

Number of GC Clause 14

Termination

In the event of termination of the contract due to any reason as already defined in the General Conditions of Contract, the Bidder shall be responsible for providing to the Authority the Goods till the time of alternate arrangements.

Number of GC Clause 16

Conflict of Interest:

The Procuring Agency reserves the right to determine on a case-by-case basis whether the Bidder should be disqualified from providing goods or services due to a conflict of a nature described in Clause GCC 17.

Number of GC Clause 20

Liquidated Damages

If the Bidder fails to provide services as required under the contract or in case of any data loss/data breach or any incident compromising the data security or other such failures related to any services, the Bidder shall pay to the Procuring Agency as Liquidated Damages at a rate of **0.27%** to **10.00%** of the Contract value, in accordance with the extent of performance failure & the cost of investigating such incidents as judged by the Authority.

Number of GC Clause 21

Performance Guarantee:

The amount of performance guarantee shall be **10.00%** of the contract price in acceptable form of **Pay Order, Call at Deposit, Demand Draft**

Number of GC Clause 27

Currency of Payment:

All the payment to be released to the contractor/Bidder shall be in Pakistani Rupees.

Number of GC Clause 28

Payment terms:

Payment will be made to the Bidder against the procured Goods and services according to the actual invoice or running bills submitted by the Bidder against the services provided within the time given in the conditions of the contract.

Number of GC Clause 29

Identifying Defects:

The Authority reserves the right at any time to inspect the premises of the provider to inspect the goods and monitor the goods being provided.

Inspections & Tests Requirements

For being Brand New, bearing relevant reference numbers of the equipment (Certificate from supplier)

For Physical Fitness having No Damages (Certificate from supplier)

For the Country of Origin as quoted by the Supplier (Certificate from manufacturer)

For conformance to specifications and performance parameters, through Prior to delivery inspection (Inspection Report by Procurement Committee / Inspection Team)

For successful operation at site after complete installation, testing and commissioning of the equipment (Installation, Testing and Commissioning Report by Procurement Committee / Inspection Team)